BTS-III(S)-05.17-0750 Reg.No. [ ] C

B.Tech. Degree III Semester Supplementary Examination May 2017

SCHOOL OF ENGINEERING, KOCHI-682 022
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
LIBRARY

ME 15-1306 MACHINE DRAWING
(2015 Scheme)

Time : 4 Hours Maximum Marks : 60

(Missing dimensions if any may be assumed)

I. (a) Sketch the internal and external ACME thread profile of nominal size of 25 mm and pitch of 3 mm. (5)

(b) Draw a Lewis foundation bolt $\Phi 20\,mm$ and indicate all the proportions on the drawing. (10)

OR

II. Draw the sectional view from the front, and view from the side of a cotter joint with sleeve used to connect two rods of 30 mm diameter each. (15)

III. Draw (i) half sectional view from the front, top half in section and (ii) view from the side of a bushed pin type flange coupling (figure 1), indicating proportions to connect two shafts, each of diameter 40 mm. (20)

OR

IV. Draw the following views of a Foot step bearing (figure 2), suitable for supporting a shaft of diameter 50 mm: (i) half sectional view from the front, with right half in section and (ii) top view. (20)

V. Figure 3 shows the details of steam stop valve. Assemble and draw (i) sectional view from the front, and (ii) top view. (25)

OR

VI. Figure 4 shows the details of steam engine stuffing box. Assemble the parts and draw (i) half sectional view from the front and (ii) top view. (25)

(P.T.O.)

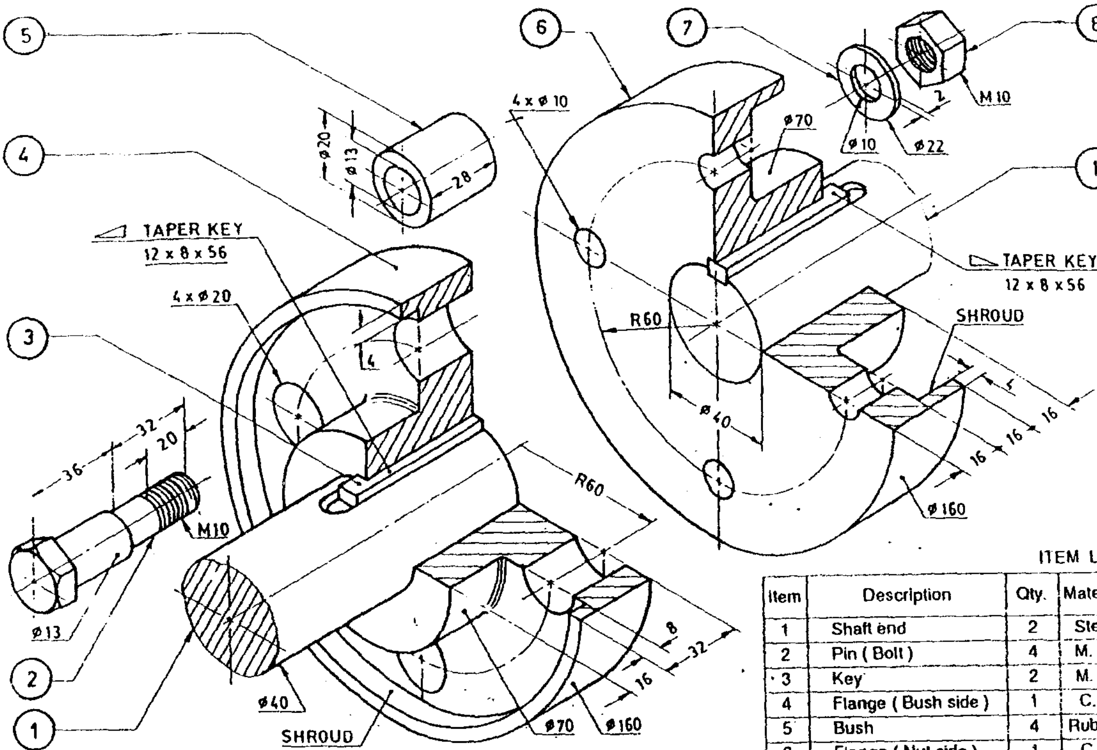


Figure 1

ITEM LIST

Item	Description	Qty.	Material
1	Shaft end	2	Steel
2	Pin (Bolt)	4	M. S.
3	Key	2	M. S.
4	Flange (Bush side)	1	C. I.
5	Bush	4	Rubber
6	Flange (Nut side)	1	C. I.
7	Washer	4	M. S.
8	Nut	4	M. S.

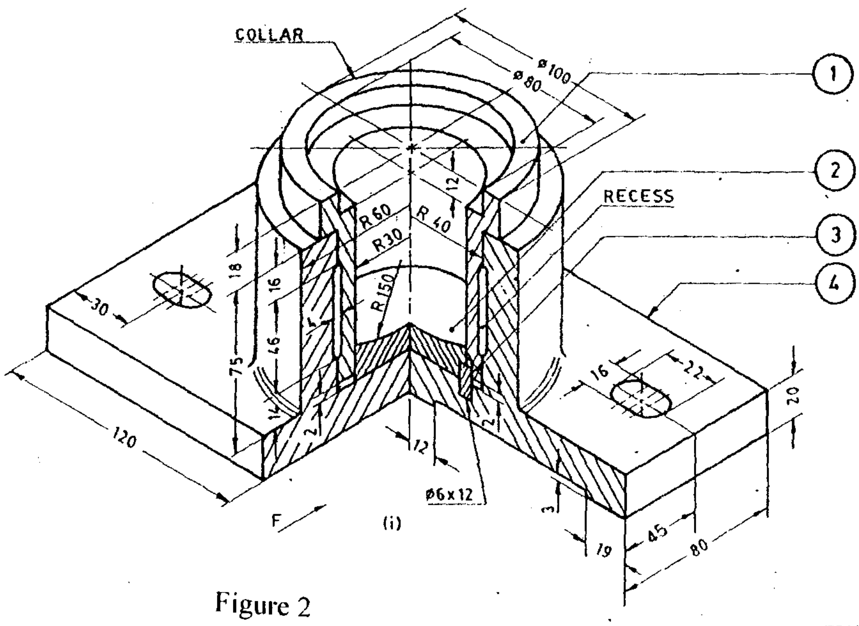
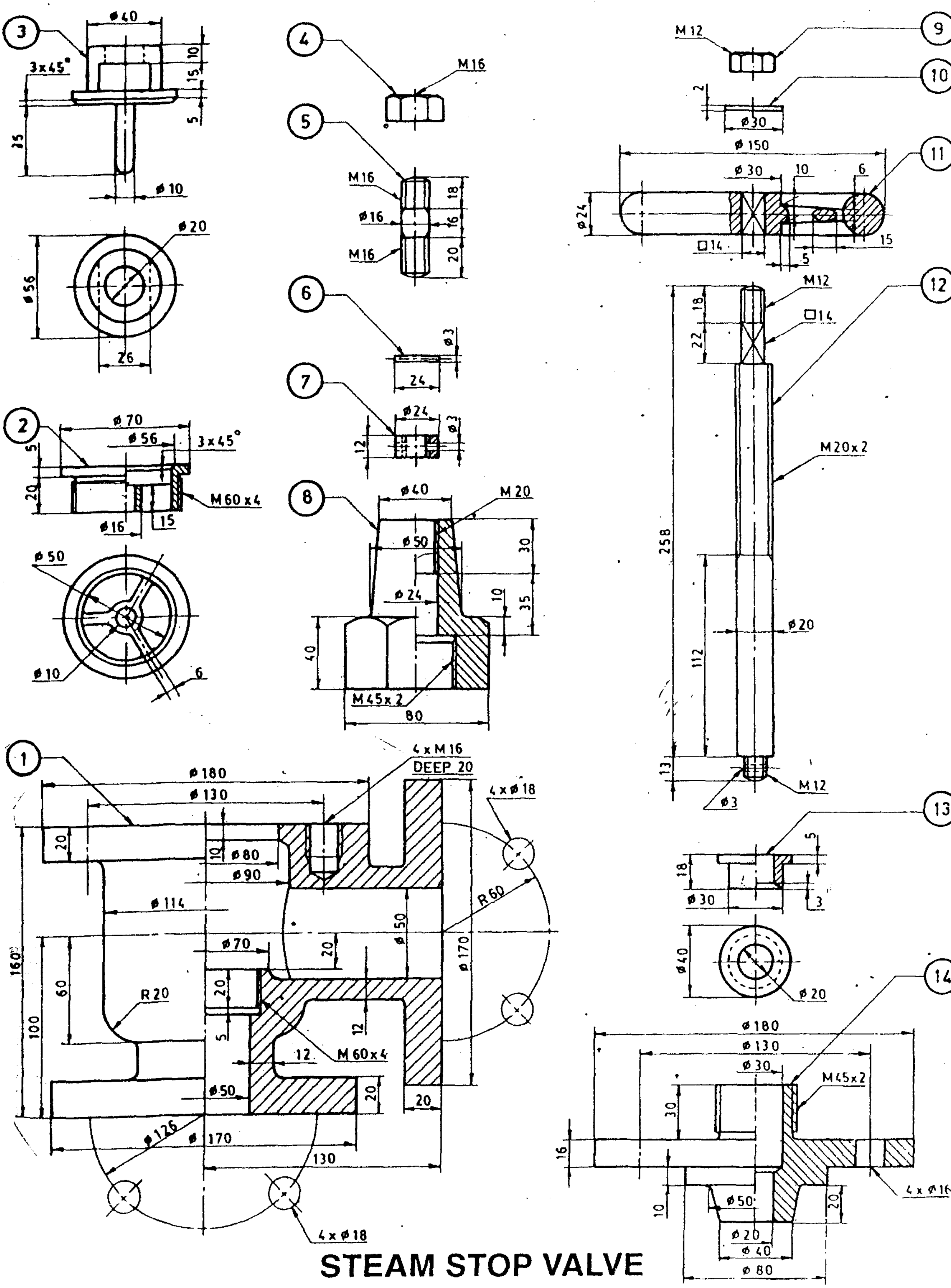


Figure 2

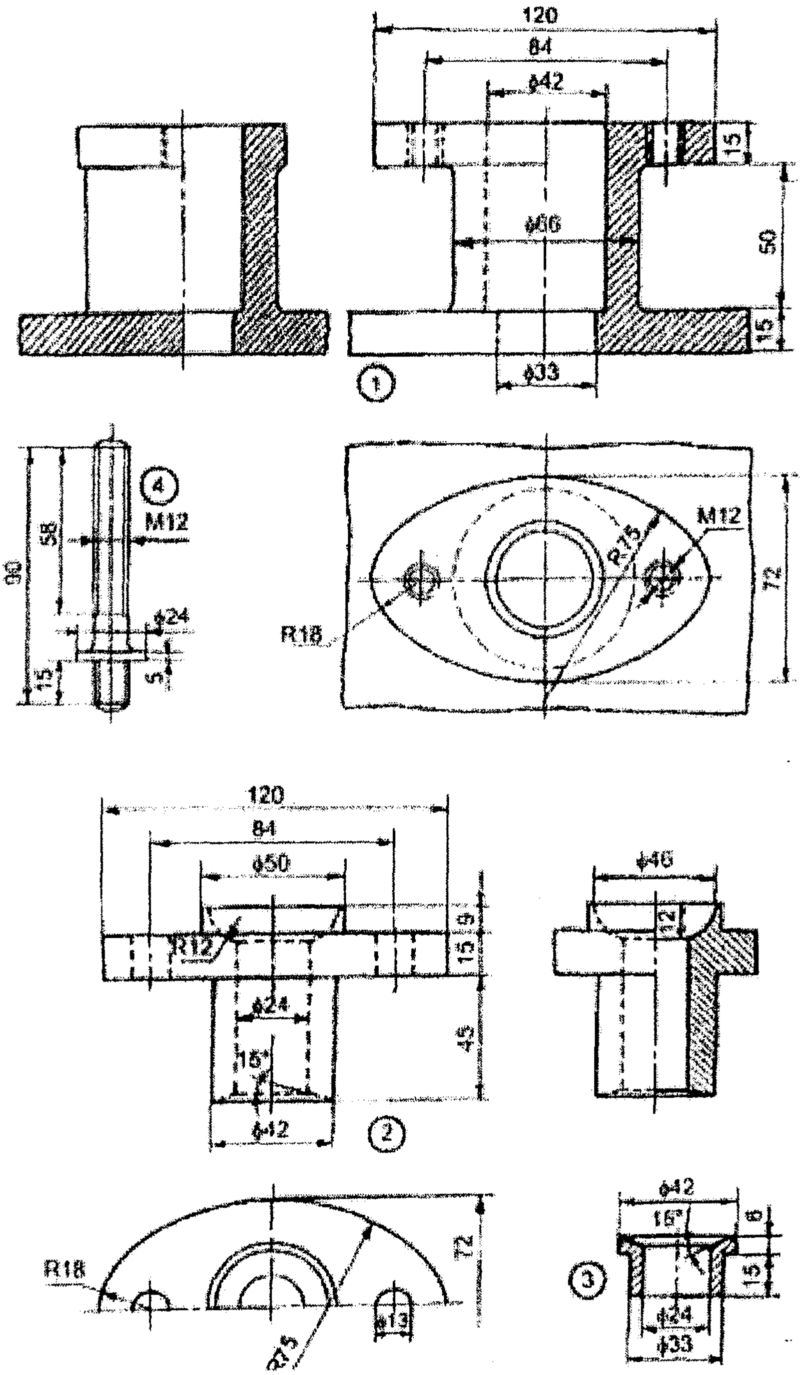
ITEM LIST

Item	Description	Qty.	Material
1	Bush	1	G. M.
2	Pad	1	G. M.
3	Snug	1	Steel
4	Block	1	C. I.



STEAM STOP VALVE

Figure 3



Parts list

Part No.	Name	Matl	Qty
1	Body	CI	1
2	Gland	Brass	1
3	Bush	Brass	1
4	Stud	MS	2
5	Nut, M12	MS	2

Figure 4

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